# 04/21/2026: Diffusion with Transformers

Reference: *Scalable Diffusion Models with Transformers*

CSCI 546: Diffusion Models

# Table of contents

1. Course Logistics
2. Lightning Chat: Diffusion with Transformers
3. Mini-Lecture: Diffusion with Transformers
4. Coding Workshop On Attention: Part II
5. Mini-Lecture: Attention

# Course Logistics

# Reading Survey #9

After having read this paper, if you were constructing a diffusion model for images, which backbone would you use, and why?

# Lightning Chat: Diffusion with Transformers

# Lightning Chat

# Mini-Lecture: Diffusion with Transformers

# Convolutional Neural Networks (CNNs)

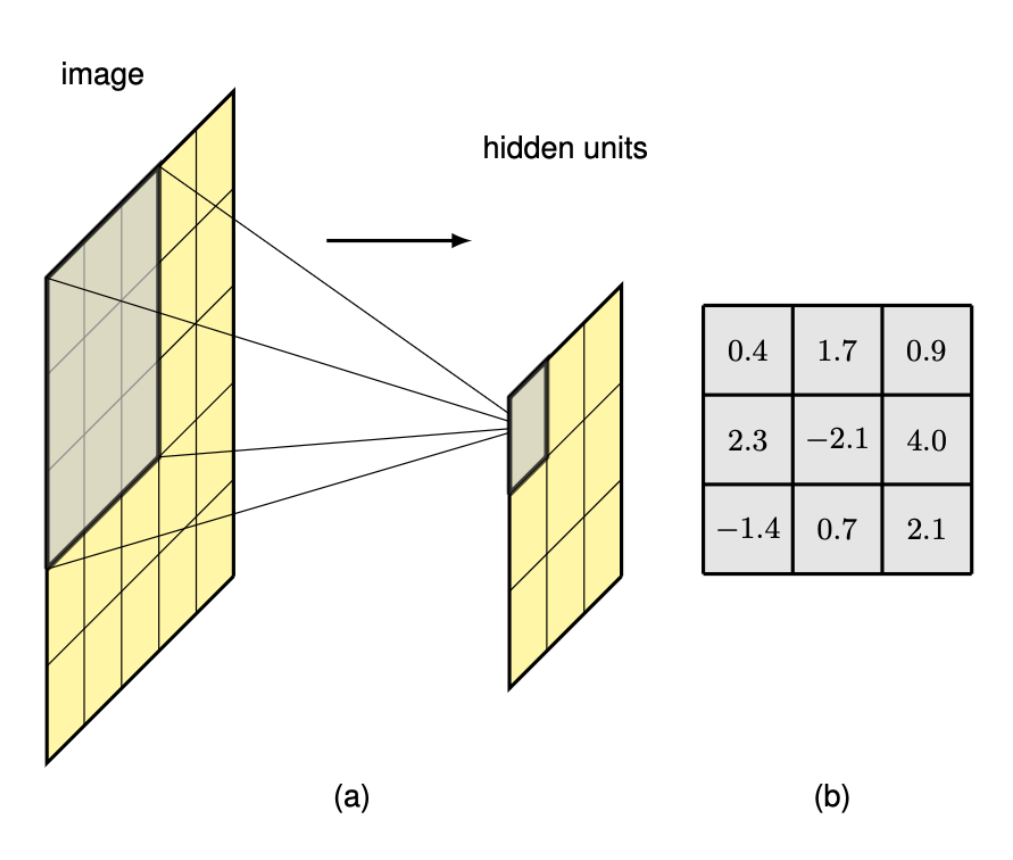



**Figure 1:** A Convolution Filter (Bishop, Deep Learning, pp.291)

# Convolutional Neural Networks (CNNs)

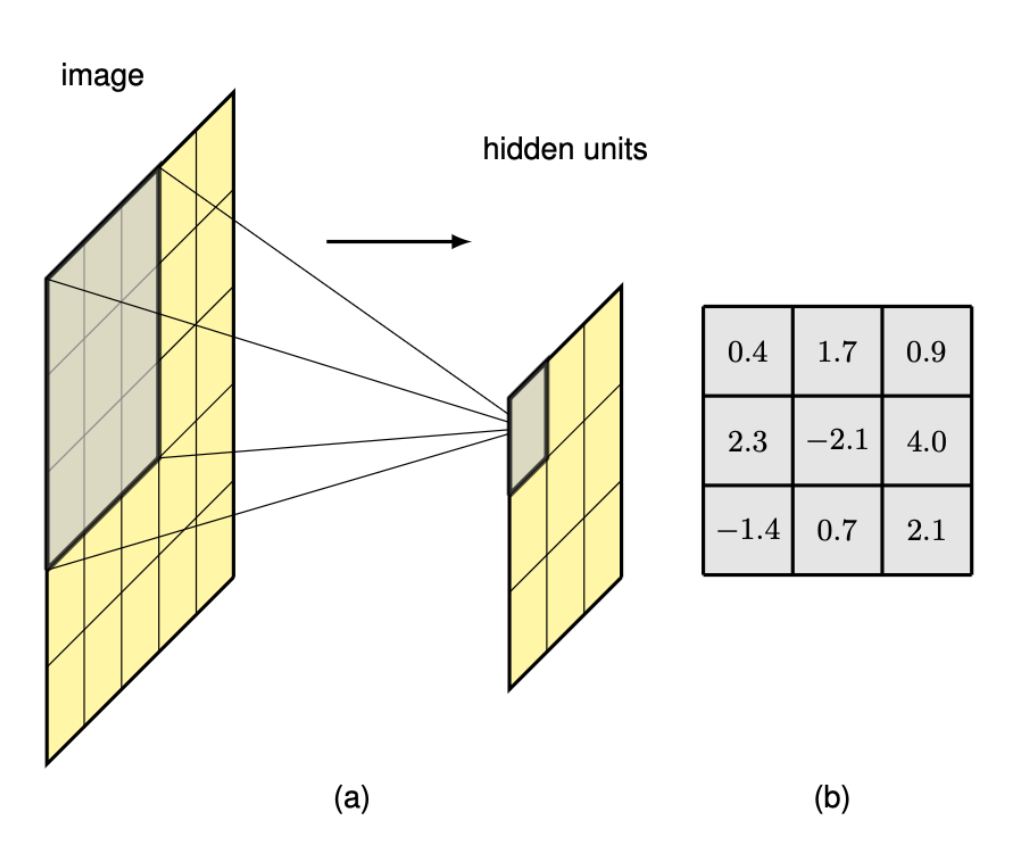



**Figure 1:** A Convolution Filter (Bishop, Deep Learning, pp.291)

**Poll.** Who can explain what's going on here?

# Convolutional Neural Networks (CNNs)

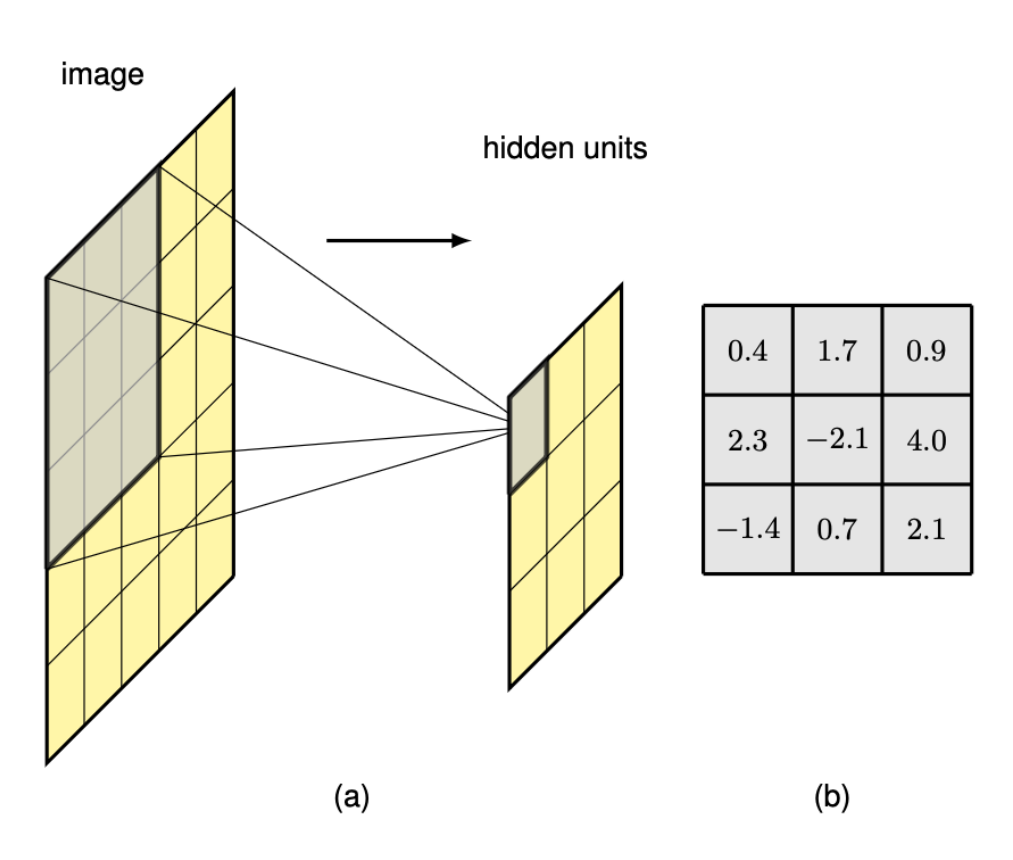
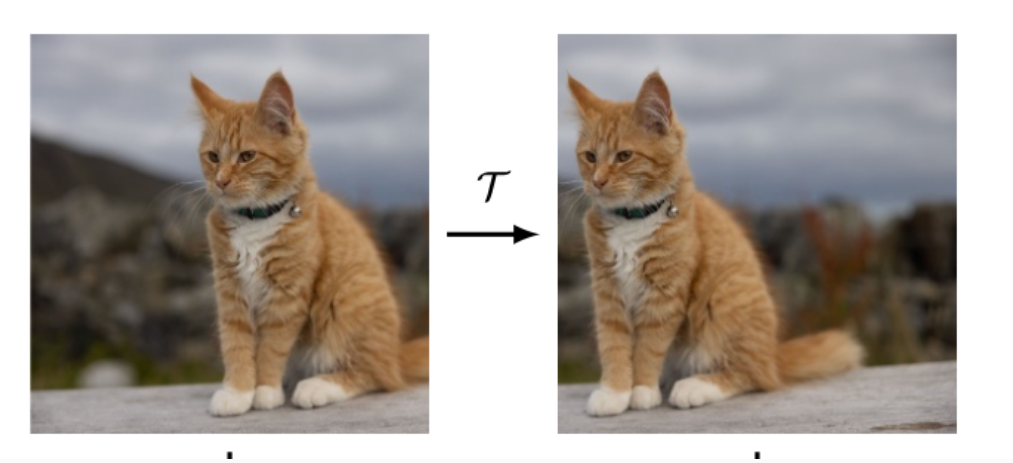



**Figure 1:** A Convolution Filter (Bishop, Deep Learning, pp.291)

**Remark.** We use the **same** hidden-unit weights at multiple locations in the image.

# Translation Invariance

The CNN has an inductive bias for translation invariance in classification.




(Bishop, Deep Learning, pp.260)

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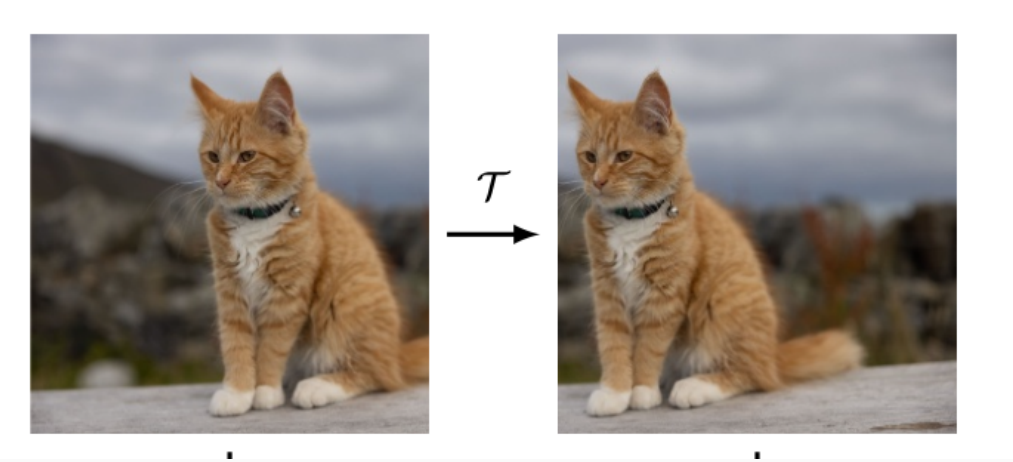


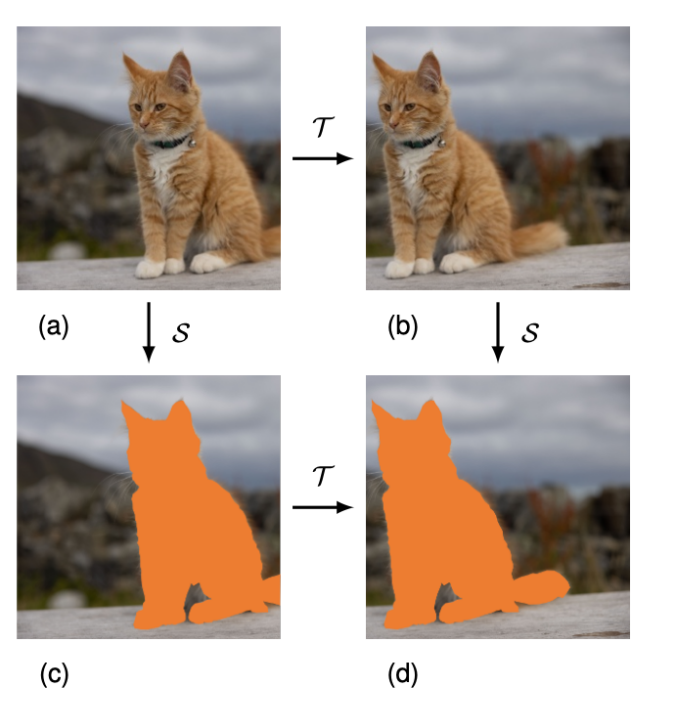

(Bishop, Deep Learning, pp.260)

If $\mathcal{C}$ is a neural network classifier, $\mathcal{T}$ is a translation operator, and $\boldsymbol{I}$ is an image, then we want

$$\mathcal{C}(\mathcal{T}(\boldsymbol{I})) = \mathcal{C}(\boldsymbol{I})$$

# Translation Equivariance

The CNN also has an inductive bias for translation equivariance for segmentation.




(Bishop, Deep Learning, pp.260)

If $\mathcal{S}$ is a segmentation network, $\mathcal{T}$ is a translation operator, and $\boldsymbol{I}$ is an image, then we want

$$\mathcal{S}(\mathcal{T}(\boldsymbol{I})) = \mathcal{T}(\mathcal{S}(\boldsymbol{I}))$$

# Vision Transformers

The surprise of vision transformers (ViT) is that the inductive biases induced by the CNN model architecture become **unnecessary** for modeling images at sufficiently large scales.

## An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale

**Alexey Dosovitskiy**[*,†], **Lucas Beyer**[*], **Alexander Kolesnikov**[*], **Dirk Weissenborn**[*], **Xiaohua Zhai**[*], **Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby**[*,†]

[*]equal technical contribution, [†]equal advising

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# Vision Transformers: How They Work

**Poll.** But transformers operate on linear 1D sequence of tokens. How do we apply it to 2D images?

Reference: An image is worth 16x16 words.

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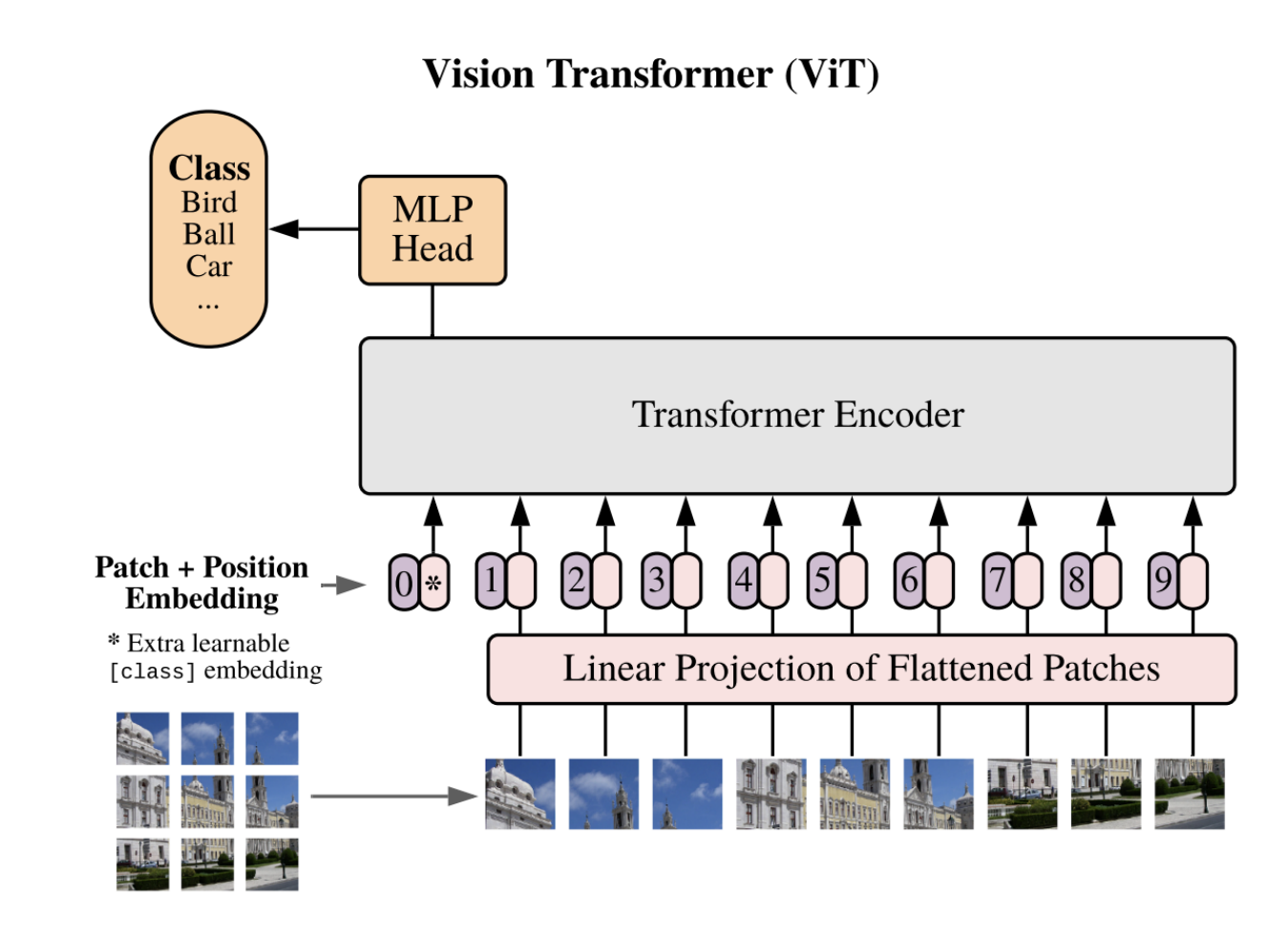


Reference: An image is worth 16x16 words.

# Vision Transformers for Classification: The Role of Attention

The model attends to regions that are semantically relevant for classification.

**Figure 2:** Representative examples of attention from the output token to the input space.

Reference: An image is worth 16x16 words.

# Vision Transformers in Diffusion

Diffusion models with ViT backbone make good generative models, as large as the model capacity is sufficiently large.

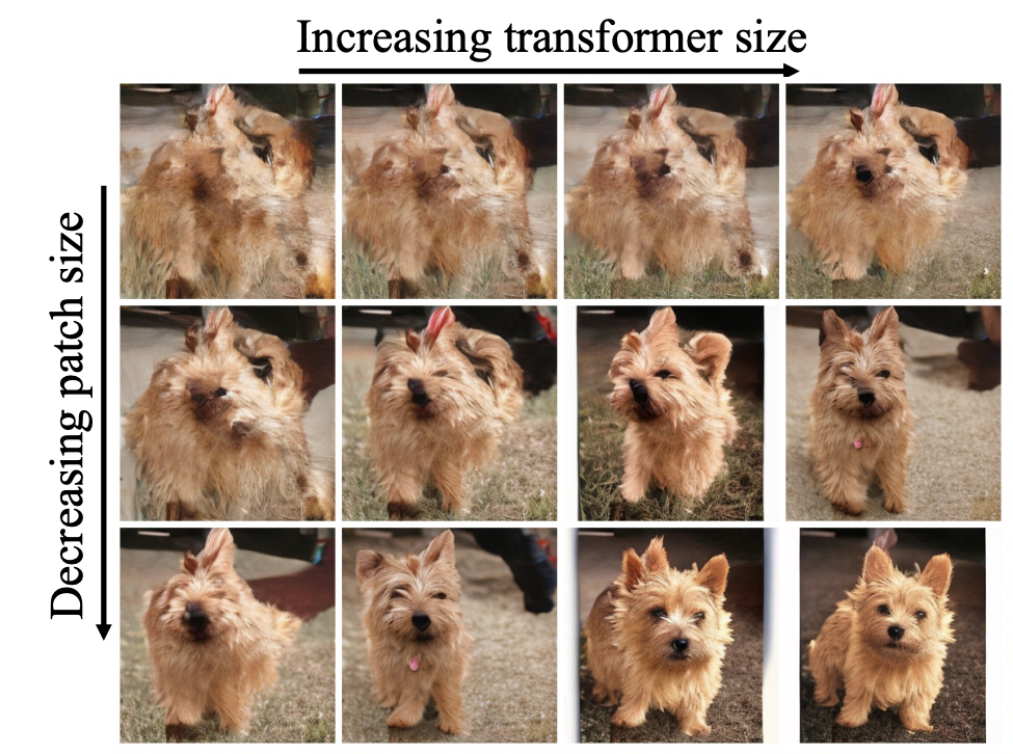


**Figure 3:** *Increasing transformer Gflops increases sample quality.* Increasing the Gflops - either by increasing transformer depth/width or increasing the number of input tokens - yields significant improvements in visual fidelity.

Reference: Scalable Diffusion Models With Transformers.

# Relevance to final project

**Poll.** So should anyone doing image generation use ViT instead of UNets as the backbone for their diffusion model?

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> *When trained on mid-sized datasets such as ImageNet [1M images], these models yield modest accuracies [...] This seemingly discouraging outcome may be expected: Transformers lack some of the inductive biases inherent to CNN's, such as translational equivariance and locality [...] However, the picture changes if the models are trained on larger datasets (14M-300M images). We find that large scale training trumps inductive bias.*
>
> — An image is worth 16x16 words.

# Final project: General leads about diffusion modeling

**Remark.** Between 2020-2023, the following led to improvements in diffusion modeling (of images):

- Predicting the noise ✓

Reference: Scalable Diffusion with Transformers, pp. 4197-4198

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- Latent diffusion models

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# Final project: General leads about diffusion modeling

**Remark.** Between 2020-2023, the following led to improvements in diffusion modeling (of images):

- Predicting the noise ✓
- Latent diffusion models
- Classifier-free guidance
- Cascaded diffusion models

Reference: Scalable Diffusion with Transformers, pp. 4197-4198

# Coding Workshop On Attention: Part II

# Complete Coding Workshop

- Sam and Dillon
- Owen
- Matt

# Mini-Lecture: Attention

**See previous slide deck**